

MATH 101 – Introduction to Mathematical Analysis I – Fall 2024

It is the prerogative of the Math 101 team to change the course during the term at our discretion. Any changes will be communicated via Blackboard, so be sure to check it frequently. Course announcements, assessment information, and other details will be regularly posted to Blackboard. You are expected to be fully aware of the following policies and expectations, so review this information carefully and ask your instructor if you have further questions.

Instructor: Peter Mao

Office Hours: Tuesday 3:00-5:00 pm and Wednesday 2:00-3:00 pm at Korman 207

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Text: Finite Mathematics and Calculus with Applications, Second Custom Edition for Drexel University, Pearson. ISBN: 978-0-136-586272
Purchasing a hard copy of the textbook is optional since a digital copy comes with MyLab (see below).

Required

Materials: There are three **required materials** for the course:

1. A MyLab (ML) access code. This code will allow access to the online assignments and includes a digital version of the textbook.

There are two different ways to get this code:

- (i) Buy the textbook package new from the university bookstore. It includes an access code.
- (ii) Buy the access code through the publisher's website.

Always access MyLab from Blackboard.

Pearson support <https://support.pearson.com/getsupport/s/>

2. A scientific calculator. One such option is the TI-30XS Multiview, but any basic scientific calculator is acceptable. **Graphing calculators, cell phones, smart watches, etc. will NOT be permitted on quizzes and exams under ANY circumstances. Any attempt to use such a device will be considered a violation of Drexel's academic honesty policies.**
3. A laptop or tablet. Students will take quizzes on their own laptop or tablet. These assessments will be held in class using MyLab.

Classroom Safety

If a student cannot attend class due to an illness or needs to quarantine, they should contact their instructor. This is the link with the latest Drexel updates of COVID guidelines: <https://drexel.edu/health-wellbeing/covid-19/overview/>. Make sure to follow these guidelines.

Recording of Class Activities

In general, students and others should not record course interactions and course activities in lecture, lab, studio, or recitation.

Students who have approved accommodation from the Office of Disability Resources to record online lectures and discussions for note taking purposes should inform their course instructor(s) of their approved accommodation in advance. The recording of lectures and discussions may only be carried out by the students enrolled in the class who have approved accommodation from Disability Resources with their instructors' prior knowledge and consent. Students with approved accommodations may be asked to turn off their recorder if confidential or personal information is presented.

If a student has any comments, concerns, or questions about provided class materials and/or recording, talk to your course instructor first. If this does not resolve the issue, you can also reach out to the Department Head, and use the process described for a grade appeal to move your concern forward. The process described for grade appeals can be found at: <https://drexel.edu/provost/policies-calendars/policies/grade-appeals/>

Appropriate Use of Course Materials

It is important to recognize that some or all of the course materials provided to you may be the intellectual property of Drexel University, the course instructor, or others. Use of this intellectual property is governed by Drexel University policies, including the IT-1 policy found at: <https://drexel.edu/it/about/policies/policies/01-Acceptable-Use/>

Briefly, this policy states that **all course materials, including recordings**, provided by the course instructor may not be copied, reproduced, distributed, or re-posted, unless otherwise given prior written approval by the University. Doing so may be considered a breach of this policy and will be investigated and addressed as possible academic dishonesty, among other potential violations. Improper use of such materials may also constitute a violation of the University's Code of Conduct and will be investigated as such.

Course Description and Learning Outcomes

The course introduces students to a variety of topics in mathematics involving functions, finance, and systems of linear equations with an emphasis on applications. Upon completion, the following skills are expected:

Functions

- Understand the definition of a function.
- Use functional notation, including that of composite functions.
- Graph basic functions, including piecewise functions.
- Determine the asymptotic behavior of a function.
- Know the properties of lines, polynomials, rational functions, exponentials, and logarithms.

Equation Solving

- Solve systems of linear equations by different methods.
- Solve quadratic, exponential, and logarithmic equations explicitly.

Financial Concepts

- Know the definitions of revenue, profit, and cost and how they are related to one another, including how to find the break-even point.
- Know the difference between simple and compound interest and understand for what types of scenarios each is used.
- Understand what an annuity is and where they are commonly encountered.
- Calculate the payment required to be made for a loan and the payment to be received from an investment.

MyLab

MyLab (ML) is a web-based online homework, quiz, and test system. The code to access this system is included with the purchase of a new textbook or can be purchased separate of the text. You can sign up for a 14-day free trial to start on course work immediately. **Not having/purchasing an access code means failing the course.**

There will be approximately one ML homework assignment for each of the sections covered, and you will have approximately one week to complete each assignment. Assignments will be due at 11:59 PM Eastern Time on Wednesday nights, but other than this final submission deadline for each assignment you will have an unlimited number of attempts. **It is your responsibility to make sure that assignments are completed by their due dates, so make sure that you do not wait until the last minute to attempt them.** Late submissions will be accepted only within one week after the original due date, and any questions answered after the due date will earn a maximum score of 50%. Exception: The last ML assignment will be due Sunday, Dec. 8 and no late submissions will be accepted for it. For each past due assignment a score of 0 will be given at the end of the course.

Practice Problems

It is extremely unlikely for you to be successful in the course if your only practice is completing the ML assignments. For each section covered in class we have listed additional practice problems (found on the last page of the syllabus) which have been chosen to illustrate important concepts and techniques that you are expected to master. These problems should be worked in detail; they will not be turned in or graded.

Quizzes

There will be 6 quizzes, and these will be 25 minutes each and given on the Thursdays of weeks 2, 3, 5, 6, 7, and 9. Quizzes will be held in class and will be done on MyLab using your own laptop or tablet. **Make sure to bring a fully charged laptop or tablet to class on the day of each quiz.** There is a “no make-up” quiz policy for this course. The lowest quiz grade will be dropped while calculating your Quiz Average. One of the reasons we drop the lowest quiz grade is in case a student is unable to attend due to unforeseen events or serious circumstances.

Exams

There will be two 50-minute midterm exams given in-class. The dates of these exams are

Exam 1: Thursday, October 17 (Week 4)

Exam 2: Thursday, November 14 (Week 8)

An announcement about the coverage and location of each exam will be given by instructors at least one week prior to the exam. A single make-up exam covering the material on **both** midterms will be offered during the last week of the course to students who miss one of the midterm exams with a valid excuse. Anyone missing a midterm exam who fails to take the make-up exam will be given a score of 0 for the exam. Note that the make-up **may not** count for both midterms, so if both midterm exams are missed you will receive a 0 for one of the exams. **Exams may not be retaken for any reason.**

The final exam is not cumulative: it will cover material from chapters 2 and 5 and will be given during finals week (Monday, Dec. 9 – Saturday, Dec. 14); the exact date and time will be determined by the University Registrar and will be announced during the term. **Travel plans are not a legitimate reason for missing the final exam. THE FINAL EXAM WILL NOT BE GIVEN EARLY FOR ANY REASON.**

Extra Help

You can receive extra assistance in the Math Resource Center (MRC): it is staffed by faculty and teaching assistants who can help with your math courses.

MRC website: <https://drexel.edu/coas/academics/departments-centers/mathematics/math-resource-center/>

Peer Mentors

The Center Learning and Academic Success Services (CLASS) offers both peer and professional academic coaching. Academic coaches work with students on any topic related to being successful academically, including study skills, time management, navigating Drexel, organization skills, test-taking, etc. Students will be encouraged to set goals and work towards accomplishing them through participation in a combination of individual coaching, identification and utilization of resources, and targeted action planning.” Students interested in working with a peer academic coach can schedule through DrexelOne:

- Log in to DrexelOne using your Drexel user ID and password, then select the "Academics" tab and follow the link to Accademia in the "Academic Support" section.
- Go to "Appointments" and "Wizard," select "Center for Learning and Academic Success" at the center, indicate which days you are available, and select "PEER" as the subject area.

If you have any questions about making an appointment, please contact CLASS at academicsuccess@drexel.edu. More information is available at <https://drexel.edu/studentlife/student-success/academic-support/center-learning-academic-success/coaching>

Course Grade

Your course grade will be computed as follows:

ML Homework Average	22%
Quiz Average	18%
Exam 1	20%
Exam 2	20%
Final	20%

Course grades will be rounded to the nearest whole number.

Letter grades are based on the course grade as follows:

A+	97% - 100%	B+	87% - 89%	C+	77% - 79%	D+	65% - 69%
A	93% - 96%	B	83% - 86%	C	73% - 76%	D	60% - 64%
A-	90% - 92%	B-	80% - 82%	C-	70% - 72%	F	< 60%

There will be no curving of any grades, and there will NOT be any extra credit opportunities either during the quarter or after the quarter is over.

Academic Honesty

Academic honesty is taken **EXTREMELY** seriously at Drexel University.

Cheating and other forms of academic misconduct are serious offenses. Anyone caught cheating on a quiz or exam for this course will earn at least a 0 on the assessment and a letter sent to the Office of Student Conduct. Further penalties may be administered.

Students should be familiar with the following policy:

<https://drexel.edu/provost/policies-calendars/policies/academic-integrity/>

Dropping and Withdrawing:

The university policies for dropping and withdrawing a course can be found at:

<https://drexel.edu/provost/policies-calendars/policies/course-add-drop/>

<https://drexel.edu/provost/policies-calendars/policies/course-coop-withdrawal/>

Disabilities and Accommodations

Students requesting accommodations due to a disability at Drexel University need to request a current Accommodations Verification Letter (AVL) in the ClockWork database **before accommodations can be made**. These requests are received by Disability Resources (DR), who then issues the AVL to the appropriate contacts. For additional information, visit the DR website at <https://drexel.edu/disability-resources/> or contact DR for more information by phone at 215.895.1401, or by email at disability@drexel.edu.

Incomplete Eligibility

To be eligible to potentially receive a grade of incomplete for the course, a student must be unable to finish the course by the end of the quarter **due to extreme circumstances that are beyond their control** (documentation may be required) and **BOTH** of the following must be true:

- 1) At least 70% of course work must be completed with a passing grade of C or better on the completed work.
- 2) A formal incomplete grade contract must be filled out and signed by both the instructor and the student before the grades are submitted for the course.

In cases where incompletes are given, only missing work that does not violate the makeup policies on the syllabus can be made up and no work may be repeated (exams and quizzes may not be retaken, and assignments may not be redone).

More details can be found at https://drexel.edu/provost/policies-calendars/policies/incomplete_grades/

Tentative Course Calendar

Always verify dates/coverage of quizzes/exams with your individual instructor.

Week	Dates	Chapters	MML HW Due (Wednesdays)	Quizzes/Exams (Thursdays)
1	Sept. 23-Sept. 29	1.1, 1.2		
2	Sept. 30-Oct. 6	10.1, 10.2	1.1, 1.2	Quiz 1 (1.1, 1.2)
3	Oct. 7-Oct. 13	12.3, 10.2.6	10.1, 10.2	Quiz 2 (10.1, 10.2)
4	Oct. 14-Oct. 20	10.3, Review 1	12.3, 10.2.6	Exam 1 (Thurs., Oct.
5	Oct. 21-Oct. 27	10.4, 10.5	10.3	Quiz 3 (10.3)
6	Oct. 28-Nov. 3	5.1, 5.2	10.4, 10.5	Quiz 4 (10.4, 10.5)
7	Nov. 4-Nov. 10	5.3, 2.1	5.1, 5.2	Quiz 5 (5.1, 5.2)
8	Nov. 11-Nov. 17	2.2, Review 2	5.3	Exam 2 (Thurs., Nov.
9	Nov. 18-Nov. 24	2.2 (cont'd), 2.3,	2.1, 2.2	Quiz 6 (2.1, 2.2)
	Tues., Nov. 26	Review class	Thanksgiving	Thanksgiving
10	Dec. 2-Dec. 8	2.5, Final Review	2.3, 2.4 (late submissions due Sunday, Dec. 8)	
Finals	Dec. 9-14		2.5 due Sunday, Dec. 8 (late submissions not accepted)	

The exact date and time of the final exam will be determined by the University Registrar and will be announced during the term. **Travel plans are not a legitimate reason for missing the final exam.**

THE FINAL EXAM WILL NOT BE GIVEN EARLY FOR ANY REASON.

Practice Problems

The following problems should be attempted after the corresponding section is taught. You do not need to submit practice problems in any form: they will not be graded; however, **it is impossible to do well in a math class without practicing**. By doing these problems you will be able to better identify the places where you may have questions.

<u>Chapter/Topic</u>	<u>Problems</u>
1.1 Slopes & Equations of Lines	1, 11, 21, 23, 25, 29, 33, 49, 61, 67(a-c)
1.2 Linear Functions and Applications	1, 5, 19, 24, 26, 28, 34
10.1 Properties of Functions	7, 9, 19, 25, 31, 35, 37, 52, 55, 73(a-g)
10.2 Quadratic Functions; Translation and Reflection	6, 8, 21, 32, 48, 52, 66
12.3 Compositions of Functions	2, 6, 10, 12, 18, 20
10.2.6 Piecewise Functions	2, 8, 18, 28, 42, 54
10.3 Polynomial & Rational Functions	7, 12, 14, 24, 26, 35, 46(a-c)
10.4 Exponential Functions	3, 11, 20, 26, 28, 45
10.5 Logarithmic Functions	5, 7, 19, 34, 49, 60, 79
5.1 Simple & Compound Interest	11, 15, 27, 31, 49, 57(a-c)
5.2 Future Value of an Annuity	19, 24, 37, 51, 57
5.3 Present Value of an Annuity	3, 7, 30, 39, 42, 43, 45
2.1 Solutions of Linear Systems Using Substitution, Elimination, & the Echelon Method	1, 9, 36, 48 (may use substitution, elimination, or echelon method)
2.2 Solutions of Linear Systems Using the Gauss-Jordan Method	4, 12, 14, 16, 19, 33, 47 (may use echelon method or Gauss-Jordan method)
2.3 Addition & Subtraction of Matrices	17, 27, 33, 39
2.4 Multiplication of Matrices	3, 11, 14, 17, 23, 47
2.5 Matrix Inverses	2, 7, 14, 16, 28, 31, 46, 47